

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

LAB 04 — INTRODUCTION TO SUPERVISORY CONTROL AND DATA ACQUISITION (SCADA)

Student Edition

Authorized ICDFA training environments only
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Lab 04: Introduction to Supervisory Control and Data Acquisition (SCADA)

Course: Intro to Critical Infrastructure for IT Learners | **Estimated Duration:** 3–4 hours

Submission deadline: As published in ICDFA E-Campus.

Authorization & Safety: Perform all activities only on the ICDFA-provided virtual machines, packet captures and isolated training networks. Do not bridge these VMs to production networks, scan external systems, or reuse credentials observed in training data.

Learning Outcomes

- Explain the purpose and architecture of SCADA systems.
- Identify the roles of PLCs/RTUs, HMI, alarms, event recording and historization.
- Start FUXA in the training SCADA VM and load a prepared HMI project.
- Connect a SCADA HMI to the OpenPLC Runtime over Modbus TCP in the isolated lab.
- Observe an automated tank-fill process and interpret operator-facing indications.

Laboratory Environment

System	Expected Address	Account	Password
PLC VM	192.168.1.10	plc	password
SCADA VM	192.168.2.10	scada	password

Note: If the instructor-provided VM image uses updated addressing or credentials, use the values published in E-Campus. Do not alter addressing unless instructed.

Pre-Lab Checklist

- Read the complete lab before beginning.
- Confirm you can access the required VM(s) and that virtualization support is enabled.
- Use the instructor-specified isolated/host-only network.
- Create or restore the clean starting snapshot for this lab.
- Prepare a report document with your name, registration number, course, lab number and date.
- Capture evidence as you progress rather than repeating the lab at the end.

Procedure

Step 1. Import/start the ICDFA PLC and SCADA VMs using the isolated network configuration supplied by your instructor. Create LAB04-START snapshots if they are not already available.

Step 2. On the SCADA VM, sign in and open a terminal. Start FUXA using the training image command: `sudo fuxa`. Enter the SCADA account password when prompted.

Step 3. Open Firefox and browse to <http://127.0.0.1:1881/editor> to access the local FUXA editor.

Step 4. Use the Save/Open Project function to load TankFillTest.json from the instructor-provided project location.

Step 5. Open FUXA project settings > Connections. Edit the openplc device and set the Modbus TCP endpoint to 192.168.1.10:502. Save the connection.

Step 6. Switch to the PLC VM. Open <http://localhost:8080/> in the local browser and sign in to OpenPLC Runtime using the credentials configured in the training image.

Step 7. Navigate to Programs, select the provided TankFillSim.st program, upload it, name it Tank Fill Sim and allow it to compile.

Step 8. Go to Dashboard and start the PLC. Confirm that the runtime status shows Running.

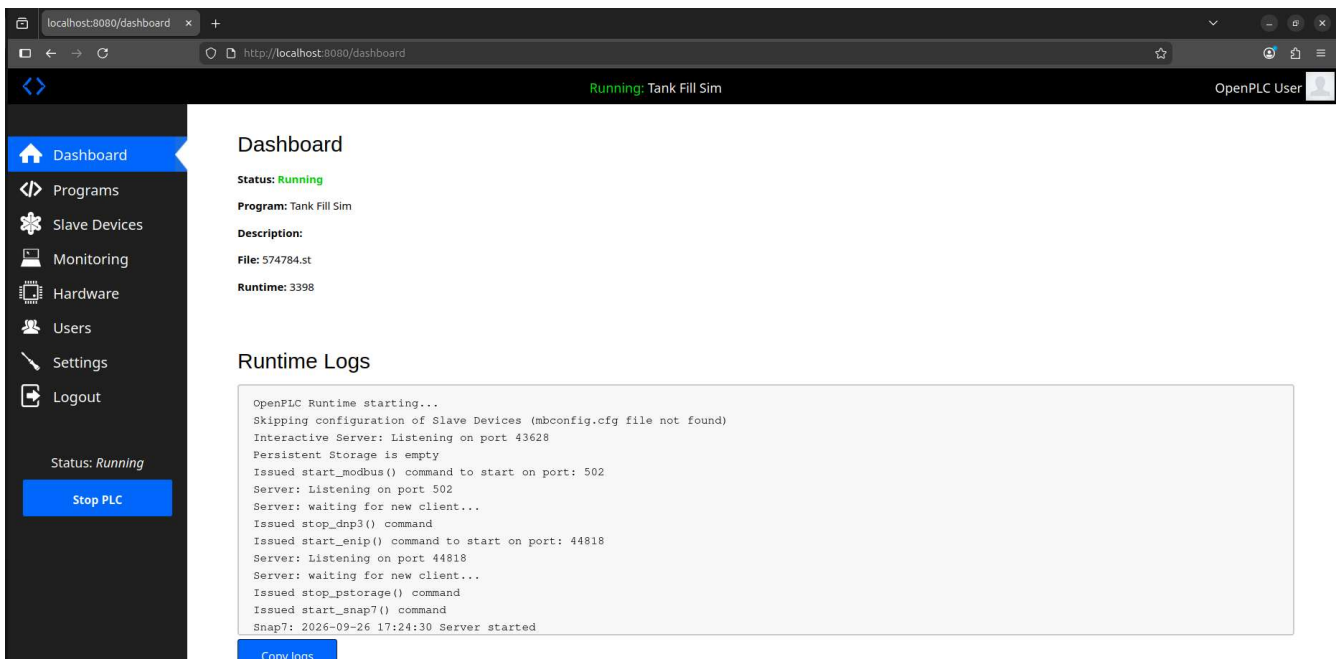
Step 9. Return to the SCADA VM and FUXA. Run/Play the HMI project.

Step 10. Observe the tank-fill simulation. Record the tank-level range, valve open/closed indication and any trend information presented to the operator.

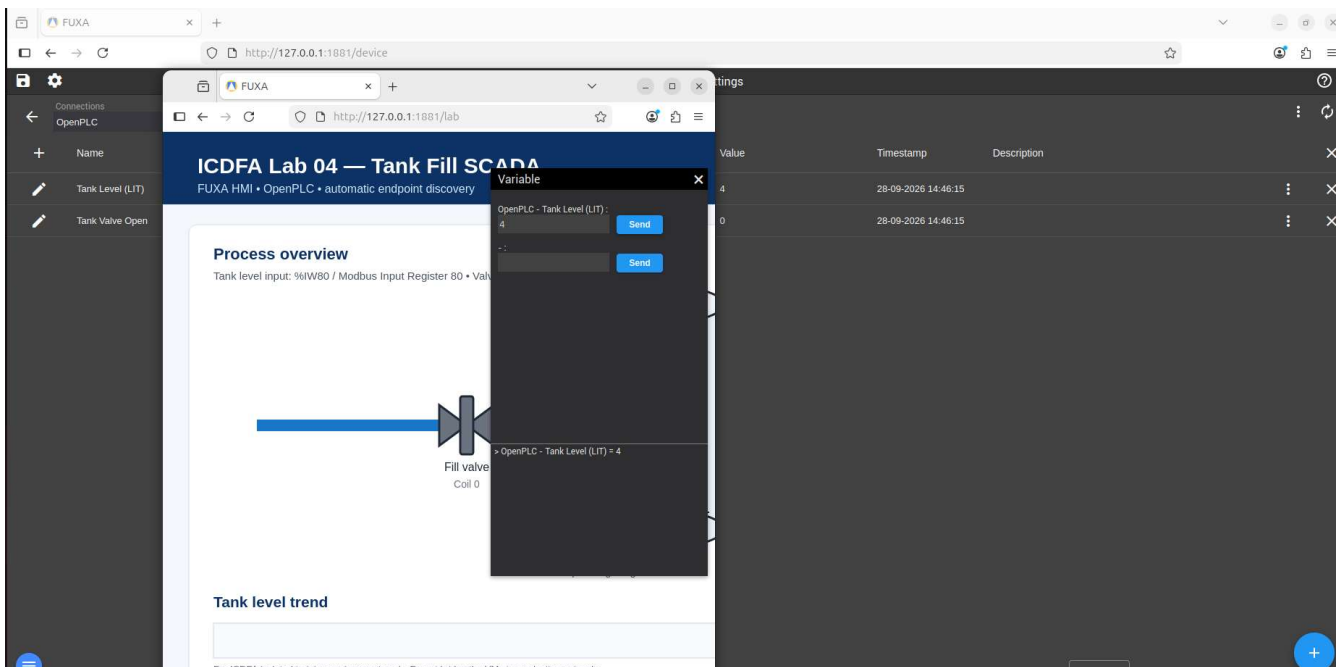
Step 11. Create a simple architecture diagram in your report showing Field/Process -> PLC -> Industrial Protocol/Network -> SCADA/HMI -> Operator, and indicate where alarms/events/historian functions belong conceptually.

Step 12. Capture the required screenshots and stop the simulation cleanly when finished.

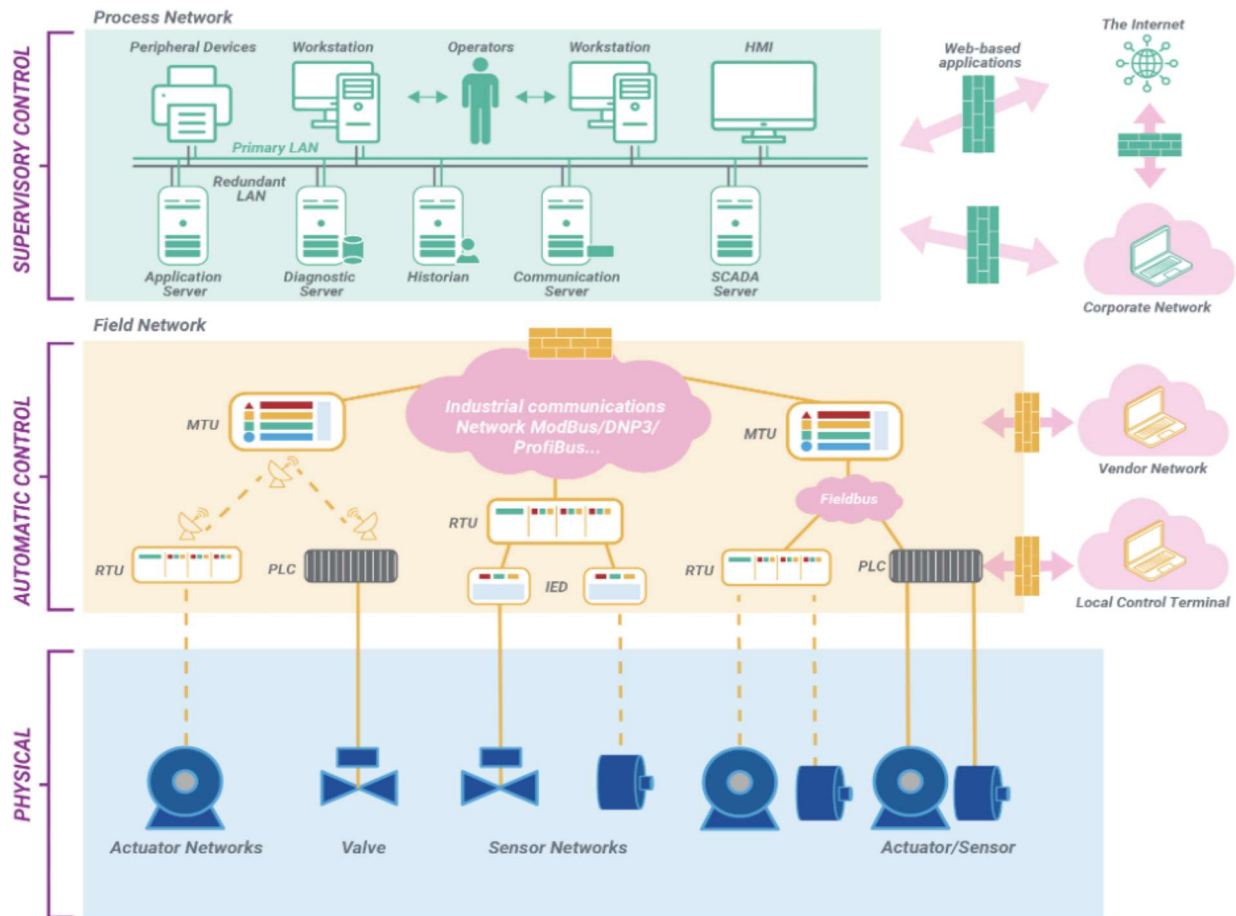
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IMG: OpenPLC Runtime in Running state.



IMG: FUXA tank-fill HMI in operation.



IMG: Labelled SCADA architecture diagram.

The difference between control logic in the PLC and supervisory visualization in the HMI/SCADA layer.

At the heart of industrial automation lies a fundamental division of labor between real-time process control and human supervision. The PLC acts as the deterministic execution engine of the factory floor. Operating on tight scan cycles measured in milliseconds, the controller reads raw electrical signals from field sensors, runs its programmed logic, and directly toggles physical outputs like valves, motors, and relays. It handles hard-wired safety interlocks and critical closed-loop control with absolute predictability. Crucially, the PLC runs independently; even if the network fails or every operator screen goes black, the PLC continues executing its logic to keep the physical process running safely.

In contrast, the HMI and SCADA layer serves as the visual window into that underlying control system for human operators. Rather than manipulating hardware directly, the SCADA system periodically polls the PLC for data, translating raw memory registers into intuitive graphic displays, animated pipelines, historical trend graphs, and active alarm lists. Operators use the HMI to monitor overall plant health, adjust operational setpoints, or issue high-level commands, which are then passed down to the PLC for execution. While the PLC makes autonomous, split-second decisions to control physical machinery, HMI and SCADA provide the situational awareness and data management needed for human operators to supervise the system effectively.

Required Evidence & Submission

- Screenshot showing OpenPLC Runtime in Running state.
- Screenshot showing the FUXA tank-fill HMI in operation.
- A labelled SCADA architecture diagram.
- A 150–250 word explanation of the difference between control logic in the PLC and supervisory visualization in the HMI/SCADA layer.

Recommended filename:

<RegistrationNo>_LAB04_Introduction_to_Supervisory_Control_and_Data_Acquisition_SCADA.pdf

Knowledge Check

1. What is the primary purpose of an HMI in a SCADA system?

Response: HMI is a user interface such as a screen or dashboard on a device that displays data and monitors the I/O of industrial processes.

2. What is the difference between an alarm, an event record and historized process data?

Response: Alarm alerts operators to potential problems with a process while an event record captures actions by both operators and automation. Historized process data are data stored and processed over time for trending and predictive analysis of a process, they are also used to create compliance reports in regulated industries.

3. Why should a SCADA operator have context rather than only raw sensor values?

Response: Adding context rather than only raw sensor values enables SCADA operators to immediately grasp the state of the process, spot abnormal behavior, and make fast, informed safety decisions instead of guessing what a raw integer means.

4. What role does the PLC play compared with the SCADA application?

Response: The PLC functions as the dedicated, real-time control engine running direct input/output logic to keep physical processes safe and responsive. The SCADA application sits a layer above as a supervisory platform, aggregating data across PLCs to handle visualization, centralized operator commands, long-term historical logging, and alarm management.

Marking Rubric — 100 Marks

Criterion	Marks	Standard
Technical execution	50	Lab procedure completed correctly; required configuration/analysis works as intended.
Evidence	20	Screenshots/files are complete, readable, relevant and clearly labelled.
Analysis & knowledge check	20	Responses demonstrate understanding of the industrial-control concepts observed.
Professional reporting	10	Clear structure, correct terminology, concise writing, file naming and submission compliance.
Submission Integrity: Evidence must come from the student's own assigned laboratory environment. Screenshots should show enough context to verify the activity. Do not submit another learner's screenshots or project files.		